

SIGNA SENSE – An Integrated Assistive Solution for the Deaf and Blind

By

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Department of Information Technology

Institute of Computer Science

A Report Submitted to the Institute of Computer Science for the Study Leading to a Project in Partial
Fulfillment of the Requirements for the Award of the Bachelor of Science Degree in Information and
Technology of Mbarara University of Science and Technology.

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II. Abstract

People with dual sensory impairments (deafness and blindness) face significant barriers to communication and safe mobility, particularly in low-resource settings where assistive technologies are scarce, expensive, or difficult to maintain. This project presents **SignaSense**, an integrated assistive solution designed to support both communication and navigation using affordable, locally obtainable electronic components.

SignaSense comprises two main modules. The first is a smart glove that captures hand gestures through bend (flex) sensing and motion sensing, processes the captured patterns on a microcontroller, and outputs the corresponding spoken message through an audio interface. The second module is a smart walking stick that uses distance sensing to detect obstacles in the user’s path and provides real-time audio alerts to support safer navigation.

The work covers system requirements, hardware design, embedded firmware development, and prototype integration. Functional testing was carried out to confirm gesture recognition responsiveness, correctness of the speech output mapping, and obstacle detection reliability at practical walking distances. The results demonstrate that combining sign-to-speech translation and obstacle-alert navigation into one solution can reduce device fragmentation and improve user independence. The project provides a foundation for future improvements such as expanded gesture vocabularies, improved robustness in varied environments, and extended user-centered evaluations.

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List of Acronyms

ICS	Institute of Computer Science
MUST	Mbarara University of Science and Technology
PWD	Persons with Disabilities
SDGs	Sustainable Development Goals
UBOS	Uganda Bureau of Statistics
WHO	World Health Organization

1 CHAPTER ONE

1.1 Introduction

In general, people with sensory disabilities face profound challenges that hinder their ability to communicate, navigate safely, and fully participate in social and economic life. According to the World Health Organization (World Health Organization 2025), more than 430 million people live with disabling hearing loss, while more than 43 million individuals are blind. A significant majority of these individuals approximately 90% reside in low- and middle-income countries where access to specialized support is often limited. These impairments often result in social isolation, reduced educational attainment, and restricted economic opportunities.

In Uganda, the challenge is further intensified by the scarcity and high cost of assistive technologies. Data from the Uganda Bureau of Statistics indicate a prevalence of disability of 12.4%, yet the local market remains dominated by expensive imported devices that typically address only one disability. This leaves a critical technological gap for individuals facing dual sensory impairments (both deafness and blindness). Furthermore, many modern assistive solutions depend on stable internet connectivity and high power consumption, making them impractical in rural or low-resource communities.

To address these barriers, this research introduces **SignaSense**, a dual-assistive system designed to promote autonomy and inclusion. SignaSense integrates two essential functions into a single platform: a smart glove that translates sign language gestures into audible speech using flex and motion sensors and a smart walking stick that utilizes ultrasonic sensors to provide real-time audio feedback for obstacle detection. By prioritizing functionality and using locally accessible hardware, this project provides a contextually relevant low-cost solution that aligns with Sustainable Development Goals 3, 8 and 10 and the Ugandan Persons with Disabilities Act (2020).

1.2 Background

Assistive technologies for communication and mobility have advanced significantly; however, many solutions are designed and priced for high-income contexts. In Uganda, limited availability of specialized devices, reliance on imports, and inconsistent power and internet access constrain adoption. An integrated, low-cost device that supports both communication (for the deaf) and navigation (for the blind) can reduce device fragmentation and improve usability for users with combined sensory challenges.

1.3 Problem Statement

People with combined hearing and visual impairments in low-resource settings face limited access to affordable and locally maintainable assistive technologies for communication and safe mobility. Existing solutions are often expensive, address only a single impairment, or depend on stable internet and power, thereby limiting independence and participation in daily activities.

1.4 Objectives

1.4.1 Main Objective

To enhance communication and mobility assistance for the deaf and visually impaired.

1.4.2 Specific Objectives

1. To examine existing systems that enable communication and mobility assistance for deaf and visually impaired.
2. To design and develop a low-cost wearable system that enables communication and mobility assistance for deaf and visually impaired.
3. To test and validate the efficiency of the developed low-cost wearable system that enables communication and mobility assistance for deaf and visually impaired.

1.5 Research Questions

1. What are the strength and gaps in the existing systems that enable communication and mobility assistance for deaf and visually impaired?
2. How do we design and develop a low-cost wearable system that enables communication and mobility assistance for deaf and visually impaired?
3. How accurate is the developed low-cost wearable system that enables communication and mobility assistance for deaf and visually impaired?

1.6 Significance of the Study

The study contributes an integrated approach to assistive technology by combining communication and mobility support in one solution. Practically, it may enhance independence, safety, and social inclusion for persons with disabilities and provide a foundation for future improvements and localized production.

1.7 Scope of the Study

This project focuses on the design and development of a prototype comprising a gesture-to-speech glove and an obstacle-detection walking stick with audio feedback. The scope covers hardware design, embedded software development, and basic usability testing. Large-scale clinical validation and long-term field deployment are outside the scope of this work.

- Contradict your methods - *i.e.* they should not imply methodological goals or standards of measurement, proof or generalisability of findings that the methods cannot sustain.

Scope

Physical scope – This is a brief description of the physical area of application where your project will apply...say, a University, a Particular business, Organization – their size or the component within such institutions where your project will apply, say Library department etc.

Technical scope – The extent of technical tools you hope to use in the project, (both languages and platforms/tools) say PHP, SQL, MySQL, HTML, Frontpage, Java, C or any softwares that you will use.

Time scope: this may refer to two things: the time the researcher is likely to take to complete the project from the time of the proposal to the final presentation. It may also refer to the “age” the records in years that the researcher is going to consider eg you may want to visit the records of a company which are as old as 10 years for you to be able to make good analysis, meaning that the problem is as old as 10 years and so it’s worth looking into.

Methodology

This should clearly explain the methodology and the steps a student will carry out to achieve the project objectives. It also involves scientific and technical methods used by I.T/C.S and C.E to achieve other objectives. For example: if we had the following objectives:

This should clearly explain the methodology and the steps a student will carry out to achieve the project objectives. It also involves scientific and technical methods used by I.T/C.S and C.E to achieve other objectives. For example: if we had the following objectives:

- To find out the existing systems (feasibility study) - which data collection method best suits this objective? Interviews? Questionnaires? Participation? A combination of all these?
- To discover the weaknesses of the existing system - which data collection method best suits this objective? Interviews? Questionnaires? Participation? A combination of all these?
- To analyze the existing data and information security measures/practices- which data collection method best suits this objective? Interviews? Questionnaires? Participation? A combination of all these?
- To design a computerized system that will take advantage of the weakness of the existing system. –use the SDLC? And achieve it by vb and sql? Php and mysql?

2 Salutation

This project proposal is respectfully submitted to the Faculty of Computing and Informatics, Mbarara University of Science and Technology, in partial fulfillment of the requirements for the course project in the Bachelor of Information Technology.

3 CHAPTER TWO: LITERATURE REVIEW

3.1 2.1 Introduction

Advancements in assistive technologies have significantly improved the lives of individuals with hearing and visual impairments. Research has explored wearable devices for sign language translation and sensor-based tools for mobility assistance, demonstrating the potential to enhance communication and independence. However, many existing solutions focus on single disabilities, rely on internet connectivity, or offer limited functionality for dynamic, real-world use.

This literature review critically examines prior studies on smart gloves for sign language recognition and walking sticks for the visually impaired, highlighting their strengths, limitations, and contextual relevance. It also identifies gaps in current knowledge, particularly in the integration of communication and mobility systems for users with dual impairments. Insights from these studies inform the design and development of the SignaSense system, emphasizing low-cost, offline, and user-centred solutions tailored to resource-limited environments.

2.2 Review of Literature Related to Objective 1: Smart Glove for Sign Language Translation

Recent research on wearable sensor gloves for sign language recognition has gained considerable attention worldwide. Ambar et al. (2020) developed a wearable glove utilizing flex and motion sensors to capture the American Sign Language (ASL) alphabet and convert it into real-time text. The system demonstrated high accuracy in gesture detection and immediate display of results; however, it was limited to static hand signs and did not support dynamic or phrase-level gestures, reducing its practical usability in natural communication contexts (Ambar et al. 2020).

Similarly, Abougarair (2022) designed a smart glove capable of translating sign language into speech output using Arduino sensors and Bluetooth connectivity. The glove exhibited fast response times and precise alphabet translation. Nonetheless, it lacked mobility integration and its phrase-level vocabulary remained constrained, limiting its practical application for comprehensive communication (Abougarair 2022).

These studies collectively highlight the potential of wearable technologies to enhance communication for deaf individuals while revealing a persistent gap: the absence of systems capable of recognizing both static and dynamic gestures, delivering phrase-level translation, and operating offline in resource-limited environments. The SignaSense glove builds upon these findings by employing flex and gyroscope sensors to recognize an expanded range of sign gestures, integrating phrase-level communication, and embedding offline audio translation for real-world usability.

2.3 Review of Literature Related to Objective 2: Smart Walking Stick for

the Blind

Mobility-assistive systems for visually impaired persons have progressed through sensor-based innovations that enhance environmental awareness. Olayiwola et al. (2025) developed a smart walking stick with voice guidance in African languages to support users in navigating obstacles. This culturally adapted system demonstrated accessibility for non-English speakers, but its functionality was limited to basic obstacle detection and did not accommodate varied terrain or advanced spatial feedback (Olayiwola et al. 2025).

Farooq et al. (2022) proposed an IoT-enabled intelligent stick that incorporated ultrasonic sensors for obstacle recognition and remote monitoring. While the system improved navigation and safety, its dependence on internet connectivity and high-power hardware rendered it impractical for low-resource settings (Farooq et al. 2022).

The SignaSense walking stick addresses these limitations by providing offline audio guidance through embedded low-power sensors, enabling usability in both rural and urban settings without reliance on network infrastructure.

2.4 Review of Literature Related to Objective 3: Integration of Communication and Mobility Systems

Few studies have attempted to integrate communication and mobility technologies for users with dual impairments. Most existing systems function independently, serving either deaf or blind individuals. This lack of convergence restricts accessibility and usability for persons with multiple disabilities.

SignaSense introduces a novel dual-assistive approach that merges gesture recognition with obstacle detection into a single platform. This integration addresses the functional gaps identified in prior studies while enhancing affordability and user convenience by reducing the need for multiple assistive devices.

2.5 Other Relevant Literature

Advances in embedded systems and low-cost sensor technologies have made the development of inclusive assistive tools increasingly feasible. Open-source hardware platforms such as Arduino and ESP32 microcontrollers support rapid prototyping and customization of wearable solutions (P. Bard, 2021). Additionally, offline voice synthesis modules and locally adapted language models improve accessibility in non-networked environments (Kumar, 2020). These technological developments reinforce the feasibility and sustainability of the SignaSense system within Uganda’s socio-economic and technological context.

2.6 General Remarks

Several themes emerge from the reviewed literature:

1. Most existing assistive systems address a single disability, neglecting users with dual sensory impairments.
2. Internet dependence and high cost remain significant barriers to accessibility in

low-income regions.

3. Limited focus has been placed on localization, particularly regarding language and cultural adaptation.

The SignaSense project directly responds to these gaps by introducing a dual-function, low-cost, offline-capable system designed for Ugandan users. It integrates the strengths of prior innovations while addressing their limitations through contextual adaptation and user-centred design.

3.2 2.7 Summary Table of Reviewed Studies

Authors	Title
Ambar et al. (2020)	Development of a Wearable Sensor Glove for Real-Time Sign Language Tran
Abougarair (2022)	Smart Glove for Sign Language Translation
Olayiwola et al. (2025)	A Smart Walking Stick with Voice Guidance in an African Language
Farooq et al. (2022)	IoT-Enabled Intelligent Stick for Visually Impaired People

4 CHAPTER 3: METHODOLOGY

4.1 3.0: Introduction

The study employs a mixed approach that combines experimental design and system development to create a dual-assistive wearable prototype for individuals with hearing and visual impairments. The methodology focuses on structured phases of design, implementation, and evaluation, ensuring that each stage contributes directly to the project objectives.

4.2 3.1 Sampling techniques

A **prototype-based experimental design** was adopted, guided by user-centred design principles. The development process integrates hardware prototyping and software programming, progressing iteratively from conceptualization to validation. Hardware components such as sensors and microcontrollers are assembled and tested for functionality, while software development is carried out using Arduino IDE and Python to handle gesture recognition, obstacle detection, and audio translation.

4.3 3.2 Study population and sample size

The population for the study includes individuals with hearing and visual impairments within Mbarara Municipality. Using purposive sampling, approximately 10–15 participants will be selected for prototype evaluation. The sample size ensures sufficient feedback on comfort, accuracy, and user interaction with the device.

4.4 3.3 Data Collection Procedure.

Research authorization will be sought from the Faculty of Computing and Informatics. Participants will receive a briefing on the project's purpose and procedures, followed by a consent process. During data collection, participants will use the prototypes under observation. Field notes, interviews, and test results will be recorded for analysis.

4.5 3.4: Data Collection Instruments.

Three main instruments will be employed to collect both qualitative and quantitative data for prototype evaluation and user feedback. These instruments ensure triangulation and enhance the reliability of the results.

3.4.1 Questionnaire

Structured questionnaires will be administered to participants after interacting with the prototype. The questions will assess usability, comfort, clarity of feedback (audio or

vibration), and perceived effectiveness of the system in addressing mobility and communication challenges. Responses will be measured using a Likert scale to enable quantitative analysis.

3.4.2 Interviews

Semi-structured interviews will be conducted with selected participants, instructors, and disability support specialists. These interviews will provide deeper insights into user experiences, preferences, and improvement suggestions for the glove and walking stick. The interviews will be guided by predetermined questions but will remain flexible to allow participants to share personal perspectives freely.

3.4.3 Participation

Participants will be directly involved in testing both components of the system under controlled conditions. Observations will focus on how effectively users interact with the glove for sign translation and the stick for obstacle detection. This participatory approach will help evaluate system accuracy, response time, and user adaptability while ensuring inclusivity in the design refinement process.

4.6 3.5 Data processing and analysis

Quantitative data will be analysed using Microsoft Excel, focusing on descriptive statistics such as accuracy percentages and mean response time. Qualitative data from interviews will be analysed thematically to identify recurring user experiences and insights. The combined results will guide system refinement and validate the project objectives.

4.7 3.6 Data presentation and interpretation

Data obtained from questionnaires, interviews, and participant observations will be organized, analysed, and presented in a clear and systematic manner. Quantitative data will be summarized using frequency tables, graphs, and charts to show patterns such as system accuracy, user satisfaction levels, and error rates during testing.

Qualitative feedback from interviews and participant observations will be interpreted thematically, highlighting user experiences, performance strengths, and areas requiring improvement. The results will then be compared against the project objectives to assess the effectiveness and usability of the *SignaSense* prototype.

4.8 3.7 Limitation of the study

The study may encounter several challenges that could influence the design, testing, and evaluation phases. Limited funding may restrict access to high-quality sensors and advanced prototyping tools. The short project duration could also constrain the time available for extended user testing and iterative refinement. Additionally, the small sample

size may limit generalization of findings to a wider population.

Technical limitations such as inconsistent sensor calibration, battery efficiency issues, and environmental interference (e.g., noise or uneven terrain) may affect prototype performance. These challenges will be mitigated by adopting a modular development approach, sourcing affordable components, and focusing on system functionality within the available resources.

5 APPENDICES

Comprises of a budget, Time framework, questionnaires, interview guide, proposed consent form, e.t.c

6 THE REPORT

Final Report includes **chapters 1-3** as in the proposal but the tense in which it is written is in reported speech or/and past continuous, the appendices and references are part of the back pages

7 CHAPTER FOUR: DATA PRESENTATION, ANALYSIS AND INTERPRETATION

7.1 4.1 Introduction

This chapter includes the presentation and analysis of data obtained from the field. The findings are coined to the purpose of the study. It presents the collected data in a more organized and summarized way. Frequency tables can be used for proper interpretation of study findings.

4.1 Presentation and interpretation of results on objective (research question 1)

4.2 Presentation and interpretation of results on objective (research question 2)

4.3 Presentation and interpretation of results on objective (research question 3)

Note: It's not a guarantee that all objectives will take the researcher in the field e.g. designing a database can't be achieved in chapter 4 but chapter 5

8 CHAPTER FIVE: SYSTEM DEVELOPMENT AND IMPLEMENTATION

This chapter presents the graphical user interfaces and the graphical database design, the existing systems, the new system, the flow charts and DFDs. The user interfaces contain various windows that will help the user to interact the system.

8.1 5.1. Requirements identified

8.1.1 5.1.1 User Requirements

It is good practice for systems analysts to involve different stake holders, at different stages of requirements gathering, so as to avoid a system failure. Therefore, different stake holders like medical personal, quality managers, directors, patients were approached and the following requirements were discovered.

8.1.2 5.1.2 Functional Requirements

Functional requirements are the intended behavior of the new system. This behavior may be expressed as services, tasks or functions the system is required to perform. Therefore the proposed system is able to In systems engineering and requirements engineering, a nonfunctional requirement is a requirement that specifies criteria that can be used to judge the operation of a system, rather than specific behaviors systems must exhibit. This should be contrasted with functional requirements that define specific behavior or functions. In general, functional requirements define what a system is supposed to *do* whereas non-functional requirements define how a system is supposed to *be*.

Functional requirements are usually in the form of "system shall do {requirement}_i", while non-functional requirements are "system shall be {requirement}_i", (Malan, Bredeyer, 2001)

Non-functional requirements are often called qualities of a system. Other terms for nonfunctional requirements are "constraints", "quality attributes", "quality goals", "quality of service requirements" and "non-behavioral requirements". Qualities, that are non-functional requirements, can be divided into two main categories:

1. Execution qualities, such as security and usability, which are observable at run time.
2. Evolution qualities, such as testability , maintainability, extensibility and scalability, which are embodied in the static structure of the software system.

8.2 5.2 System requirements (of the new system)

To be used efficiently, all computer software needs certain hardware components or other software resources to be present on a computer. Most software defines two sets of system requirements: minimum and recommended. With increasing demand for higher processing power and resources in newer versions of software, system requirements tend to increase over time

8.3 5.3 Architectural design of the new system

5.3.1 Description of the Designed System

5.3.2 Hardware Specifications

5.3.3 Software Specifications

5.3.4 Systems Architecture

5.5.5 Data Requirements

5.5.6 Data Inputs (database and tables)

5.5.7 System Flow

EER/ER Diagrams and DFDs

HIPO chart

Block Diagrams

Data Flow diagrams

System Flow chart

5.5.8 Physical database/system design

5.5.9 Data Outputs (System forms). These should be very well explained. (Remember this is Presentation and Discussion of results)

9 CHAPTER SIX: SUMMARY, CONCLUSION AND RECOMMENDATIONS

This chapter presents the summary of background context, objectives, methodology and findings. It also presents the conclusions and recommendations for managing the challenges of information flow in the new system and in the global ICT sector for further research.

9.1 6.1 Summary (optional)

In one paragraph, the researcher gives personal view on the implementation of the new system, from a personal perspective and global perspective, explaining how success can be achieved when new systems are used

9.2 6.2 Conclusion

6.2.1 Legal and Ethical concerns

The researcher can write about the legal and ethical implications of computerizing business process mostly, if human beings were involved in the process. This can be in relation to the policy in the country or the case study, the ministry.

6.2.2 Also, the researcher can conclude about the positive implications of adopting the new system to the case study and hoe the rest of the society can benefit

6.2.3 Were the objectives realized in relation to the user requirements?

6.2.4 What are the other factors that influence the proper adoption of the new systems?

9.3 6.3 Recommendations

Should the new system be adopted? If so how? The researcher should explain the different methods of deploying the application for you and recommend one of the methods for use by the case study. E.g. pilot conversion

9.4 6.4 Areas for further study

If the researcher feels that more research can be done on his research, he can recommend which areas eg is it use of biometrics to enhance security? e.t.c

10 APPENDICES

Consent form, letter of introduction, interview guide, and questionnaires should be provided in this section. The budget is not required

11 INDICES

Minimum Deployment Requirements e.g Software Requirements, Hardware Requirements, Network requirements, Technical Human Resource Requirements

Consider the example below:

Minimum Deployment Requirements

1. Software Requirements • Operating System for the server Windows NT, 2000 or above

- Operating system for the client PC Windows XP/NT
- Office suite
- Windows.NET framework 2,0
- Database Management System SQL server version 3:23.48
- Window installer 20. and 3.1

2. Hardware Requirements • At least 1 power stabaliser

- Processor Intel Pentium III or above
- Processor Speed 800MHZ or above
- 128MB RAM or above depending on the Operating System
- Disk Space 40 GB or above
- Bandwidth 100MBps
- UPS at least 3

3. Network requirements • Rj45 connectors

- Cat5 or 6
- Sever 2003 Or 2008 +1GB RAM
- Hp proliant Sever machine
- NICs
- At least Switches to cove the hospital
- Networking tool kit
- Dust blower

4. Technical Human Resource Requirements • 1 systems administrator

- 1programmer
- 1 systems analyst
- 1 computer technician

Bibliography

References

- Abougarair, A. J. (2022). “Smart glove for sign language translation”. In: *International Robotics & Automation Journal*.
- Ambar, R. et al. (2020). “Development of a Wearable Sensor Glove for Real-Time Sign Language Translation”. In: *International Journal of Advanced Computer Science and Applications* 11, pp. 102–109.
- Farooq, M. S. et al. (2022). “IoT Enabled Intelligent Stick for Visually Impaired People for Obstacle Recognition”. In: *Sensors* 22, p. 8914.
- Olayiwola, A. et al. (2025). “A smart walking stick with voice guidance in an African language for visually impaired persons”. In: *Journal of Electrical Systems and Information Technology* 12, p. 67.
- World Health Organization (2025). *Deafness and hearing loss*. URL: <https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss> (visited on 03/17/2026).

12 APPENDIX A: ASSESMENT FORM

Proposal /10

Oral Presentations /50

Report / 40

Criteria	Clarification	Maximum Score	Awarded Score
Problem Statement (Including Background, aims and objectives)		2	
Background		1	
Objectives	Statement clarity, rationale, relevance, originality, and adequacy and precision of presentation of aims and objectives.	2	
Literature review		3	
Attendance		2	

Criteria	Clarification	Maximum Score	Awarded Score
	Depth of understanding of subject matter	10	
Oral Presentation	Sufficiency and clarity of answers to questions	5	
	Effective communication skills	5	
	Professionalism in presentation, general overall impression	5	
	System functionality / data analysis	25	

Criteria	Clarification	Maximum Score	Awarded Score
Methodology (with reference to tools and processes used)	Appropriateness, adequacy, level of analysis, synthesis and design, precision of application of processes, tools and resources, logical flow, and adequacy of data used.	5	
Results (including products such as software/hardware, models of systems, etc)	Quality of results based on the methodology, appropriateness of results based on aim and objectives, adequacy of representation (charts) and documentation, identification of limits, relevant discussion, conclusion and recommendations.	30	
Write up	Proper Formatting	5	
Task Completion		5	
Efficiency		5	

13 APPENDIX B: TYPOGRAPHIC FORMAT

Cover Page • A cover page for the concept/proposal/report is to be provided (A4 size) in **Appendix E**

- The title of the report is to be on the top portion about 1cm from the top edge of the box.

Typing of the concept/proposal/report • They should be typed using computer print (Times New Roman with 12pt.) with nearletter-quality printing style.

- The title and main headings are to be in capital letters. If possible, bold scripts may be used with capitals.

- 1.5 line spacing is to be adopted.

- Each page is to be given a page number at the bottom-middle part of the page.

Paragraphing • Each main paragraph or heading is numbered with whole number.

- Sub-heading is given in decimals of the whole number of the main heading.

- Subsidiary paragraphs to the sub-heads are to be in decimal points of the sub-heads.

- Each paragraph deals with only one issue, while subsidiary issues are discussed in relevant sub-paragraphs or subsidiary paragraphs, as the case may be.

14 APPENDIX C: SUPERVISION MONITORING FORM

Mbarara University of Science and Technology

Institute of Computer Science Undergraduate Research and Projects Coordination Office

Undergraduate Research Projects Progress Monitoring Form / Report Group

No:..... Month:..... Department:.....

Project Title:

Project Supervisor:

Date: Signature:

Date of Meeting	Proposed Objective of Meeting	Agreed Objectives	Proposed Date of Next Meeting	Comments

15 APPENDIX D: RESEACH WORKPLAN

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UNDERGRADUATE RESEARCH PROJECTS WORKPLAN																	
SEMESTER 1	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Week 15	Week 16	Week 17
Activities																	
Workshop : Project proposal and Report writing, Literature Review and Referencing, Format of Assessment. (All 3rd years)																	
Project proposal Development																	
Project Proposal Submission																	
Data collection and Analysis																	
Submission of proposal Marks by Supervisors																	
Supervisors' Monthly Report 1																	
System Coding																	
Supervisors' Monthly Report 2																	
Supervisors' Monthly Report 3																	
Exams																	
SEMESTER 2																	
Workshop: Final year projects orientation, Idea generation, Concept Paper Development and																	

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16 APPENDIX E: FACE PAGE